

Project Development Phase

Date	11 July 2026
Team ID	
Project Name	Electrical Vehicle Performance & Market Analysis Dashboard
Maximum Marks	4 Marks

Data Preprocessing Steps

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the Electric Vehicle dataset into **MySQL** and **Tableau Public**.
 - Verified that all EV attributes such as Brand, Model, Top Speed, Range, Price, Charging Speed, Body Style, and Acceleration were imported successfully.
 - Confirmed that the dataset was ready for preprocessing.
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Step 2: Check for Missing Values

- Examined the dataset for null and blank values.
 - Missing values in **FastChargeSpeed**, **Acceleration**, and other numerical columns were identified.
 - Replaced missing numerical values with their respective average values to improve analysis accuracy.
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Step 3: Remove Duplicate Records

- Checked the dataset for duplicate EV models.
 - Removed duplicate records where necessary.
 - Ensured every vehicle appeared only once.
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Step 4: Validate Data Types

Verified that every attribute had the correct data type.

- Brand → String
 - Model → String
 - Body Style → String
 - Top Speed → Integer
 - Range → Integer
 - Acceleration → Decimal
 - Price in Germany → Integer
 - Price in UK → Integer
 - Fast Charge Speed → Integer
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Step 5: Data Cleaning

Performed several cleaning operations, including:

- Removed **km/h**, **km**, **sec**, **£**, and **€** symbols.
- Converted text values into numeric values.
- Corrected inconsistent formatting.
- Standardized body style and brand names.

Step 6: Create Calculated Fields

Created calculated fields including:

- Price Range (Lakhs)
- Range (Km)
- Top Speed (Km/h)
- Acceleration (Sec)
- Efficiency (Wh/Km)
- Fast Charging Speed (Km/h)

Step 7: Final Dataset

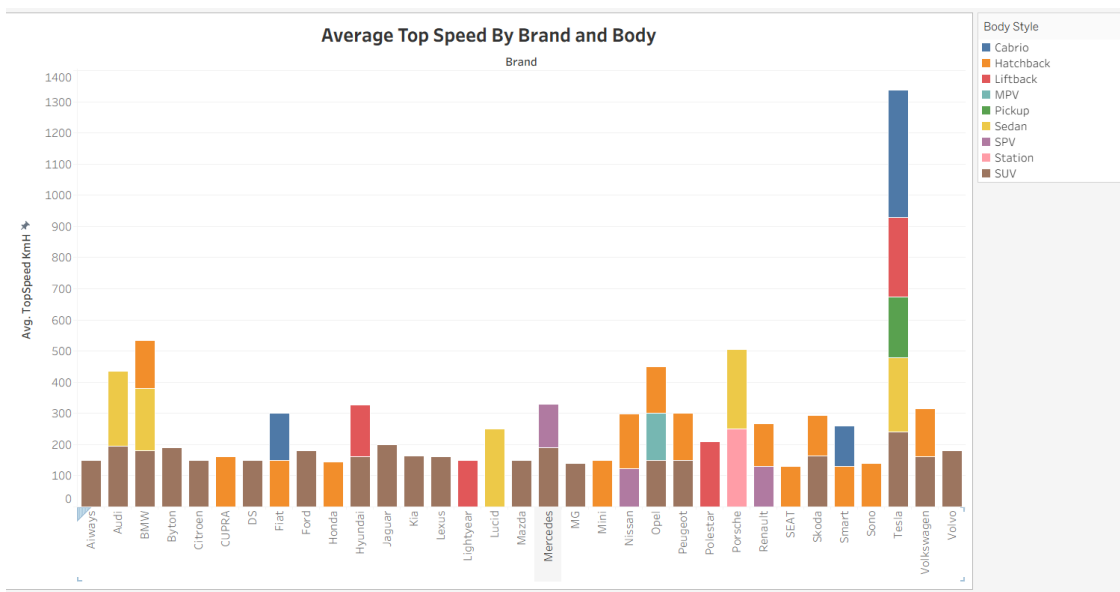
- Verified that the cleaned dataset contained no inconsistencies.
 - Confirmed that all preprocessing tasks were completed successfully.
 - Finalized the dataset for dashboard development in Tableau Public.
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Business Questions with Visualization

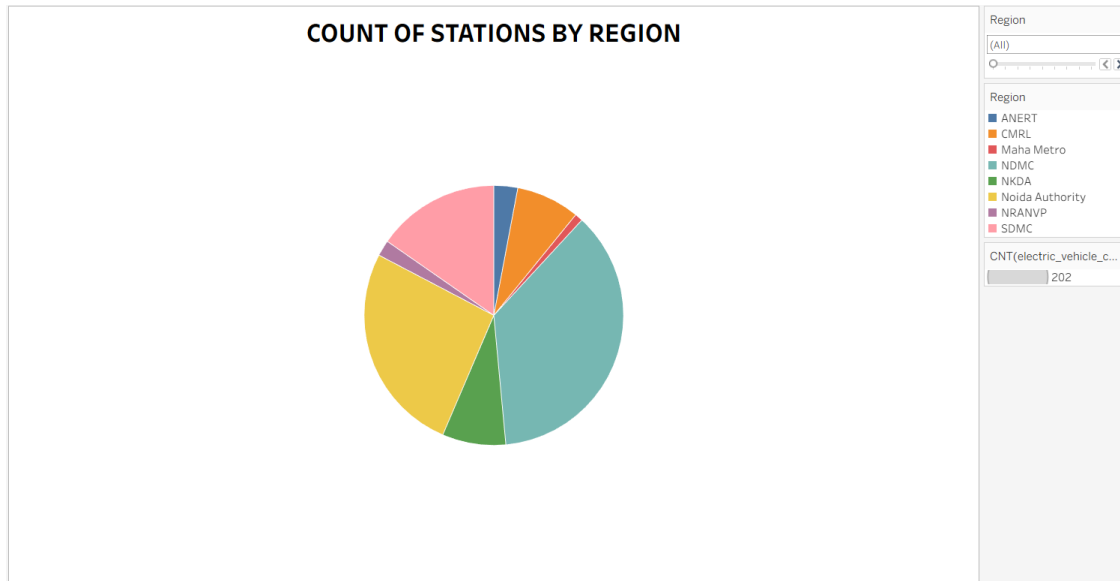
S.No	Business Question	Visualization
1	Which EV brand offers the highest average top speed?	Stacked Bar Chart
2	How are charging stations distributed across different regions?	Pie Chart
3	Which EV models have the highest top speed?	Horizontal Bar Chart
4	How does acceleration vary across body styles?	Box Plot
5	Which brands provide the maximum driving range?	Bar Chart
6	How do EV prices vary among different vehicle models?	Line Chart
7	Which body style provides better performance?	Box Plot
8	What are the overall performance insights of electric vehicles?	Interactive Dashboard

Visualizations

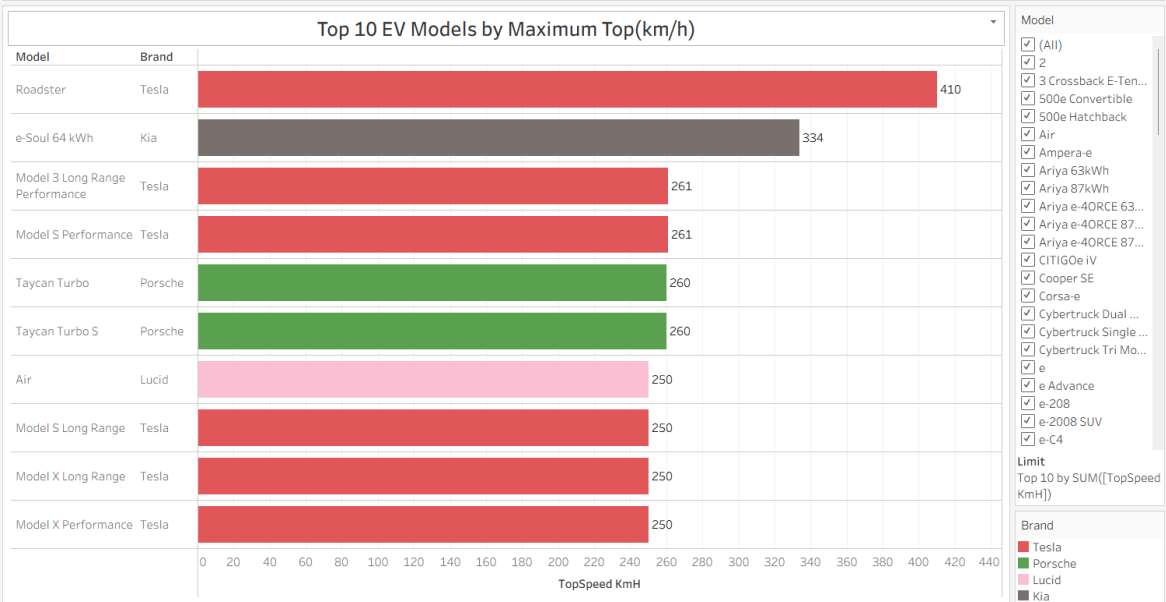
1. Average Top Speed by Brand and Body Style



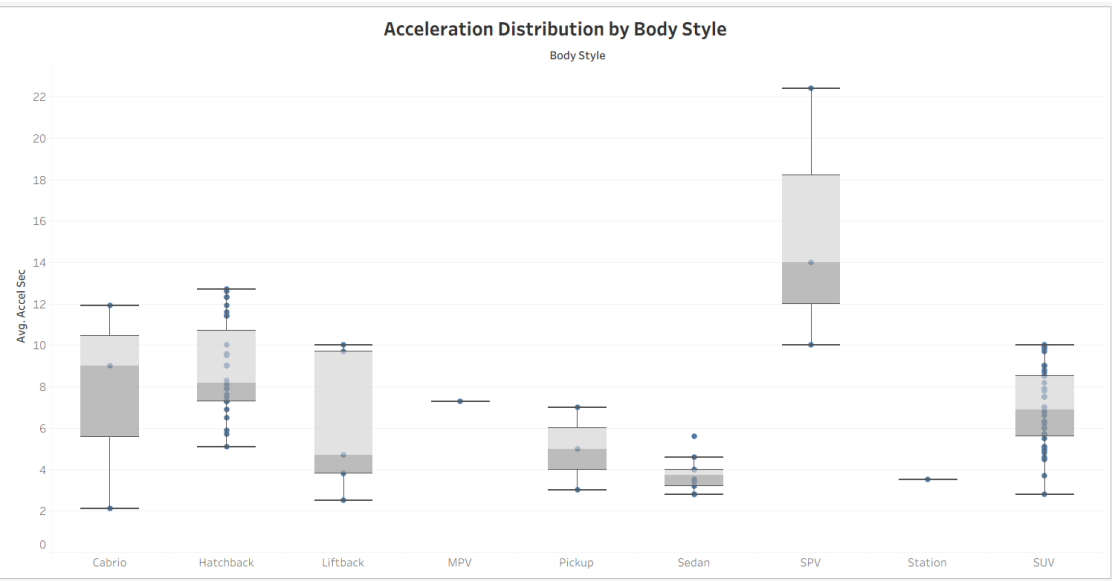
2. Count of Charging Stations by Region



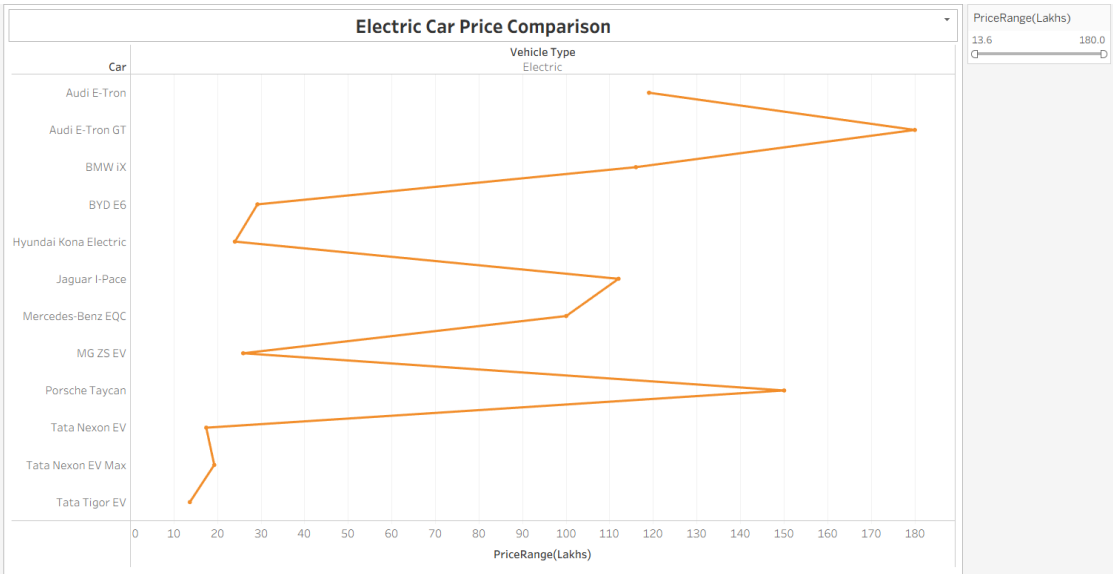
3. Top 10 EV Models by Maximum Top Speed



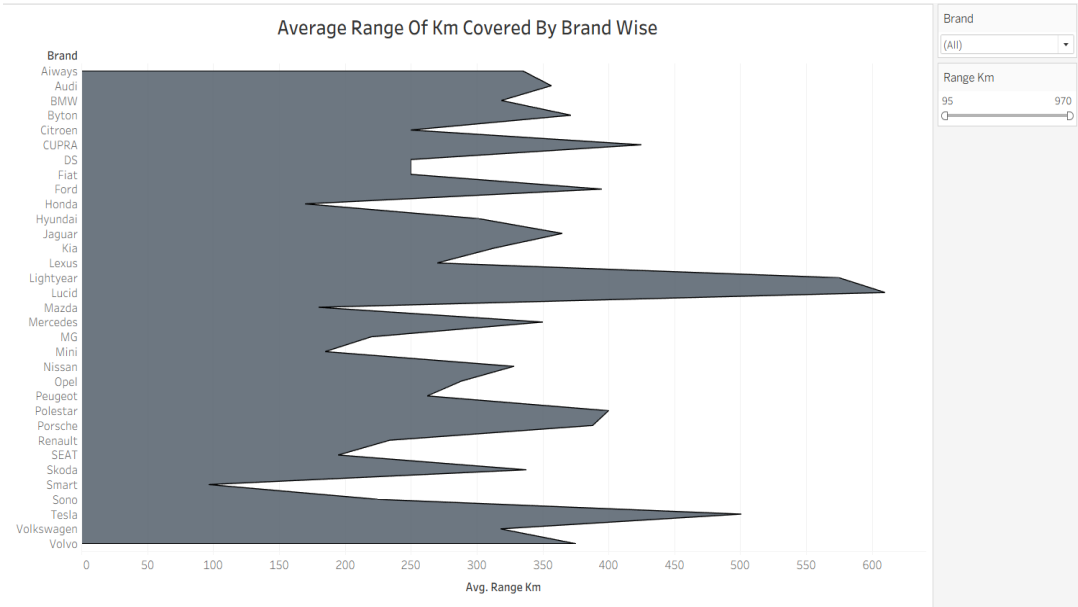
4. Acceleration Distribution by Body Style



5. Electric Vehicle Price Comparison



6. Average Range Covered by Brand



7. Final Interactive Dashboard

Electrical Vehicle Performance &Market Analysis Dashboard

